

# Introduction to Regression

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# Outline

- 1 Introduction to Regression
- 2 Simple Linear Regression (SLR)
- 3 Ordinary Least Squares (OLS) Estimation
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- 6 Estimating the Error Variance  $\sigma^2$
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# Introduction to Regression

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# What Is Regression?

Regression is a statistical technique for modeling the relationship between a **response** (dependent) variable and one or more **predictor** (independent) variables.

Instead of treating events in a system as isolated, regression lets us ask: *if this variable changes, how does that one respond, on average?*

- Response variable: what we want to explain or predict (denoted  $y$ )
- Predictor variable(s): what we use to explain it (denoted  $x$ )
- The regression equation summarizes the average relationship between them

# Why Regression?

- **Business:** does advertising spend increase sales, and by how much?
- **Medicine:** how does drug dosage affect blood pressure reduction?
- **Engineering:** how does curing temperature affect concrete strength?
- **Economics:** how does income relate to household spending?
- **Environment:** how does fertilizer amount relate to crop yield?

# Simple Linear Regression (SLR)

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# The SLR Model Equation

For an outcome variable  $Y$  and a single predictor variable  $X$ , the model is given as

$$Y = \beta_0 + \beta_1 X + \epsilon, \quad \epsilon \sim N(0, \sigma^2)$$

- $\beta_0$ : the intercept — expected or average value of  $Y$  when the value of  $X$  is zero
- $\beta_1$ : the slope — the expected or average change in sales for a one-unit increase  $X$
- $\epsilon_i$ : random error — everything affecting  $Y$  that  $X$  does not capture
- $\beta_0, \beta_1$  are unknown **population parameters**; we estimate them from data as  $\hat{\beta}_0, \hat{\beta}_1$

# The SLR Model in Terms of Observed Values

For a single predictor  $X$  (advertising spend) and response  $Y$  (sales), and observed data points  $(x_i, y_i)$ ,  $i = 1, 2, \dots, n$ , the SLR model is

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2), \quad i = 1, \dots, n.$$

The fitted line is given by

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i.$$

What are the interpretation of  $\hat{\beta}_0$  and  $\hat{\beta}_1$  in this context?

# Assumptions of the SLR Model

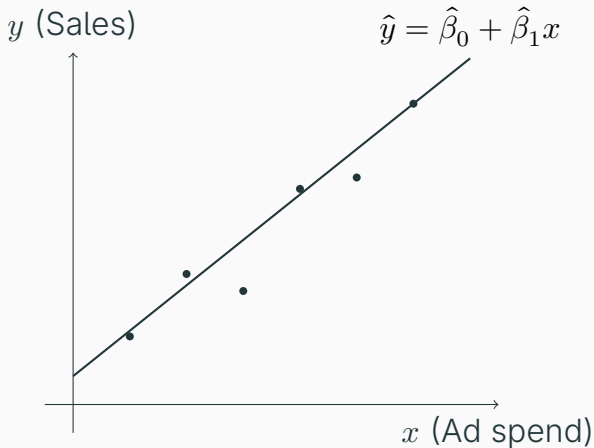
- **Linearity:** the mean of  $y$  is a linear function of  $x$ , i.e.  $E[\varepsilon_i] = 0$
- **Independence:**  $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$  for  $i \neq j$
- **Normality:**  $\varepsilon_i \sim N(0, \sigma^2)$
- **Constant variance (homoscedasticity):**  $\text{Var}(\varepsilon_i) = \sigma^2$  for all  $i$
- $x_i$  values are fixed, known constants, measured without error

# Ordinary Least Squares (OLS) Estimation

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# The Regression Line

We want a line  $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$  that fits the scatter of (spend, sales) points as closely as possible.

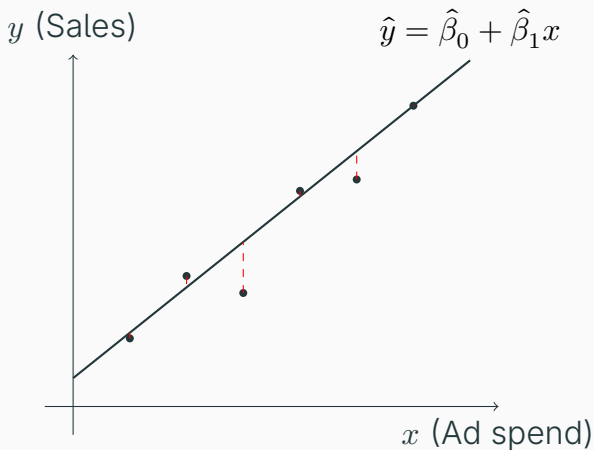


# Residuals

For each data point, the **residual** is the vertical gap between the observed  $y_i$  and the fitted value  $\hat{y}_i$ :

$$\hat{\varepsilon}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$$

Ordinary Least Squares (OLS) chooses  $\hat{\beta}_0, \hat{\beta}_1$  to make these residuals collectively as small as possible.



# Setting Up the Least Squares Problem

OLS minimizes the sum of squared residuals:

$$S(\beta_0, \beta_1) = \sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

To minimize  $S(\beta_0, \beta_1)$ , we take partial derivatives with respect to  $\beta_0$  and  $\beta_1$ , set them equal to zero, and solve. Since  $S$  is a quadratic (convex) function, this stationary point is the global minimum.

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# First-Order Conditions

Differentiate  $S(\beta_0, \beta_1) = \sum_i (y_i - \beta_0 - \beta_1 x_i)^2$ :

$$\frac{\partial S}{\partial \beta_0} = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)$$

$$\frac{\partial S}{\partial \beta_1} = -2 \sum_{i=1}^n x_i (y_i - \beta_0 - \beta_1 x_i)$$

Setting both equal to zero and evaluating at the minimizing values  $\hat{\beta}_0, \hat{\beta}_1$  gives the **normal equations**.

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# The Normal Equations

$$\sum_{i=1}^n y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^n x_i$$
$$\sum_{i=1}^n x_i y_i = \hat{\beta}_0 \sum_{i=1}^n x_i + \hat{\beta}_1 \sum_{i=1}^n x_i^2$$

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## Solving for $\hat{\beta}_1$

From the first normal equation, dividing by  $n$ :

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Substitute into the second normal equation:

$$\sum_i x_i y_i = (\bar{y} - \hat{\beta}_1 \bar{x}) \sum_i x_i + \hat{\beta}_1 \sum_i x_i^2$$

Since  $\sum_i x_i = n\bar{x}$ , this becomes

$$\begin{aligned} \sum_i x_i y_i - n\bar{x}\bar{y} &= \hat{\beta}_1 \left( \sum_i x_i^2 - n\bar{x}^2 \right) \\ \hat{\beta}_1 &= \frac{\sum_i x_i y_i - n\bar{x}\bar{y}}{\left( \sum_i x_i^2 - n\bar{x}^2 \right)} \end{aligned}$$

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## Solving for $\hat{\beta}_1$ (cont.)

Define

$$S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y}) = \sum_i x_i y_i - n\bar{x}\bar{y}, \quad S_{xx} = \sum_i (x_i - \bar{x})^2 = \sum_i x_i^2 - n\bar{x}^2$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$$

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## Solving for $\hat{\beta}_0$

Having solved for  $\hat{\beta}_1$ , we plug back into the first normal equation:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Interpretation for our example:  $\hat{\beta}_1$  is the estimated extra sales revenue per extra unit of advertising spend, and the fitted line passes through the point of averages  $(\bar{x}, \bar{y})$ .

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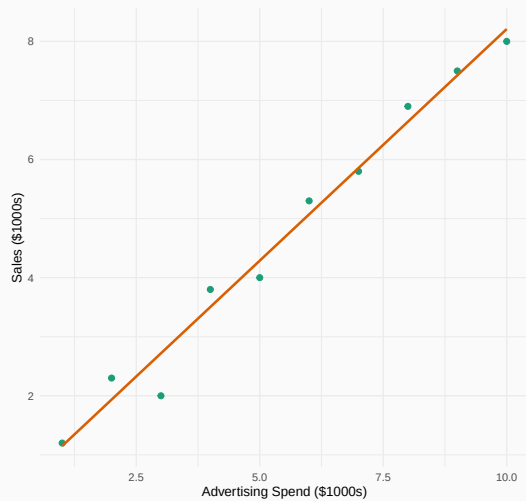
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# OLS Estimators — Summary

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \quad \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

# The Fitted Line



# Properties of the OLS Estimators

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# Rewriting $\hat{\beta}_1$ as a Linear Combination of the $y_i$

Since  $\sum_i (x_i - \bar{x})\bar{y} = 0$ , we can write

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{S_{xx}} = \frac{\sum_i (x_i - \bar{x})y_i}{S_{xx}} = \sum_{i=1}^n c_i y_i, \quad c_i = \frac{x_i - \bar{x}}{S_{xx}}$$

The weights  $c_i$  satisfy two useful properties, both provable directly from the definition of  $S_{xx}$ :

$$\sum_i c_i = 0, \quad \sum_i c_i x_i = 1, \quad \sum_i c_i^2 = \frac{1}{S_{xx}}$$

These properties are all we need to establish unbiasedness and variance.

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Treating the  $x_i$  as fixed and using  $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ :

$$\begin{aligned} E[\hat{\beta}_1] &= E \left[ \sum_i c_i y_i \right] = \sum_i c_i E[y_i] = \sum_i c_i (\beta_0 + \beta_1 x_i) \\ &= \beta_0 \underbrace{\sum_i c_i}_{=0} + \beta_1 \underbrace{\sum_i c_i x_i}_{=1} = \beta_1 \\ &\boxed{E[\hat{\beta}_1] = \beta_1} \end{aligned}$$

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# Variance of the OLS Estimators

Because the  $\varepsilon_i$  are independent with common variance  $\sigma^2$ , and  $\hat{\beta}_1 = \sum_i c_i y_i$  with  $\text{Var}(y_i) = \sigma^2$ :

$$\text{Var}(\hat{\beta}_1) = \sum_i c_i^2 \text{Var}(y_i) = \sigma^2 \sum_i c_i^2 = \frac{\sigma^2}{S_{xx}}$$

A similar calculation for  $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$  gives

$$\text{Var}(\hat{\beta}_0) = \sigma^2 \left( \frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)$$

More spread-out  $x$  values (larger  $S_{xx}$ ) give more precise slope estimates — intuitively, advertising spend that varies a lot from month to month tells us more about its effect on sales than spend that barely changes.

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# The Gauss–Markov Theorem: OLS Is BLUE

Among all estimators of  $\beta_0$  and  $\beta_1$  that are:

- **L**inear in the  $y_i$ , and
- **U**nbiased,

the OLS estimators  $\hat{\beta}_0, \hat{\beta}_1$  have the smallest possible variance.

This is why OLS is called **BLUE**: the **B**est **L**inear **U**nbiased **E**stimator. “Best” here means minimum variance, not that OLS makes no errors — it means no other linear, unbiased method can do better, under the SLR assumptions.

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## Other Useful Properties of the Least-Squares Fit

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Beyond unbiasedness and BLUE, the OLS fit has several algebraic properties that follow directly from the normal equations — worth knowing since they show up in diagnostics and in software output.

- The residuals always sum to zero:  $\sum_i \hat{\varepsilon}_i = 0$
- The observed and fitted values have the same sum:  $\sum_i y_i = \sum_i \hat{y}_i$
- The fitted line always passes through the centroid  $(\bar{x}, \bar{y})$
- Residuals weighted by  $x_i$  sum to zero:  $\sum_i x_i \hat{\varepsilon}_i = 0$
- Residuals weighted by  $\hat{y}_i$  sum to zero:  $\sum_i \hat{y}_i \hat{\varepsilon}_i = 0$

## Property 1 & 2: Residuals Sum to Zero

Recall the first normal equation from earlier:

$$\sum_i y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum_i x_i$$

Since  $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ , summing over  $i$  gives

$$\sum_i \hat{y}_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum_i x_i$$

The right-hand sides are identical, so

$$\boxed{\sum_i y_i = \sum_i \hat{y}_i} \implies \sum_i (y_i - \hat{y}_i) = \sum_i \hat{\varepsilon}_i = 0$$

This is not a coincidence — it is forced by the very equation OLS solves to find  $\hat{\beta}_0$ .

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## Property 3: The Fitted Line Passes Through $(\bar{x}, \bar{y})$

Divide the previous result  $\sum_i y_i = \sum_i \hat{y}_i$  by  $n$ :

$$\bar{y} = \bar{\hat{y}} = \frac{1}{n} \sum_i (\hat{\beta}_0 + \hat{\beta}_1 x_i) = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$$

So the point  $(\bar{x}, \bar{y})$  satisfies the fitted equation exactly — the **centroid** of the data always lies on the regression line.

This matches how we solved for  $\hat{\beta}_0$  earlier: we built the line to pass through  $(\bar{x}, \bar{y})$  by construction.

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## Property 4 & 5: Weighted Residuals Sum to Zero

Recall the second normal equation:

$$\sum_i x_i y_i = \hat{\beta}_0 \sum_i x_i + \hat{\beta}_1 \sum_i x_i^2$$

Since  $x_i \hat{y}_i = \hat{\beta}_0 x_i + \hat{\beta}_1 x_i^2$ , summing gives  $\sum_i x_i \hat{y}_i$  equal to the same right-hand side, so

$$\sum_i x_i y_i = \sum_i x_i \hat{y}_i \implies \boxed{\sum_i x_i \hat{\varepsilon}_i = 0}$$

For the last property, write  $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$  and use both results just derived:

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## Property 4 & 5: Weighted Residuals Sum to Zero

Recall the second normal equation:

$$\sum_i x_i y_i = \hat{\beta}_0 \sum_i x_i + \hat{\beta}_1 \sum_i x_i^2$$

Since  $x_i \hat{y}_i = \hat{\beta}_0 x_i + \hat{\beta}_1 x_i^2$ , summing gives  $\sum_i x_i \hat{y}_i$  equal to the same right-hand side, so

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## Estimating the Error Variance $\sigma^2$

---

# Residuals and the Error Sum of Squares

The parameter  $\sigma^2$  measures how much sales vary around the regression line for a given level of advertising spend — it is unknown and must be estimated from the residuals.

Define the **error sum of squares**:

$$SS_E = \sum_{i=1}^n \hat{\varepsilon}_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

It can be shown that  $E[SS_E] = (n - 2)\sigma^2$ . The divisor  $n - 2$  reflects that two degrees of freedom were used up in estimating  $\beta_0$  and  $\beta_1$  before we could compute residuals.

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## An Unbiased Estimator of $\sigma^2$

$$\hat{\sigma}^2 = MS_E = \frac{SS_E}{n-2}$$

Since  $E[SS_E] = (n-2)\sigma^2$ , it follows immediately that

$$E[\hat{\sigma}^2] = \sigma^2$$

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## Finding the Expected Value of $SS_E$

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## Why $E[SS_E] = (n - 2)\sigma^2$ : Rewriting the Residual

Recall  $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ . Substituting into  $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ :

$$\hat{y}_i = (\bar{y} - \hat{\beta}_1 \bar{x}) + \hat{\beta}_1 x_i = \bar{y} + \hat{\beta}_1 (x_i - \bar{x})$$

So each residual splits into a piece explained by  $x$  and a leftover piece:

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## Why $E[SS_E] = (n - 2)\sigma^2$ (cont.)

Squaring and summing over  $i$ :

$$\begin{aligned}SS_E &= \sum_i [(y_i - \bar{y}) - \hat{\beta}_1(x_i - \bar{x})]^2 \\&= \underbrace{\sum_i (y_i - \bar{y})^2}_{S_{yy}} - 2\hat{\beta}_1 \underbrace{\sum_i (x_i - \bar{x})(y_i - \bar{y})}_{S_{xy}} + \hat{\beta}_1^2 \underbrace{\sum_i (x_i - \bar{x})^2}_{S_{xx}} \\&= SS_E = S_{yy} - 2\hat{\beta}_1 S_{xy} + \hat{\beta}_1^2 S_{xx}\end{aligned}$$

# Simplifying $SS_E$

So far:

$$SS_E = S_{yy} - 2\hat{\beta}_1 S_{xy} + \hat{\beta}_1^2 S_{xx}$$

Since  $\hat{\beta}_1 = S_{xy}/S_{xx}$ , we have  $S_{xy} = \hat{\beta}_1 S_{xx}$ , so

$$\hat{\beta}_1^2 S_{xx} = \hat{\beta}_1 (\hat{\beta}_1 S_{xx}) = \hat{\beta}_1 S_{xy}$$

The last two terms combine into one:

$$-2\hat{\beta}_1 S_{xy} + \hat{\beta}_1 S_{xy} = -\hat{\beta}_1 S_{xy}$$

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## Expectation of $S_{yy}$

Subtracting  $\bar{y} = \beta_0 + \beta_1\bar{x} + \bar{\varepsilon}$  from  $y_i = \beta_0 + \beta_1x_i + \varepsilon_i$ :

$$y_i - \bar{y} = \beta_1(x_i - \bar{x}) + (\varepsilon_i - \bar{\varepsilon})$$

Squaring and summing:

$$S_{yy} = \beta_1^2 S_{xx} + 2\beta_1 \sum_i (x_i - \bar{x})(\varepsilon_i - \bar{\varepsilon}) + \sum_i (\varepsilon_i - \bar{\varepsilon})^2$$

The  $x_i$  are fixed constants and  $E[\varepsilon_i - \bar{\varepsilon}] = 0$ , so the middle term vanishes in expectation. For the last term, using  $E[\varepsilon_i^2] = \sigma^2$  and  $\text{Var}(\bar{\varepsilon}) = \sigma^2/n$ :

$$\sum_i (\varepsilon_i - \bar{\varepsilon})^2 = \sum_i \varepsilon_i^2 - n\bar{\varepsilon}^2 \Rightarrow E\left[\sum_i (\varepsilon_i - \bar{\varepsilon})^2\right] = n\sigma^2 - n \cdot \frac{\sigma^2}{n} = (n-1)\sigma^2$$

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## Expectation of $\hat{\beta}_1 S_{xy}$

Using  $S_{xy} = \hat{\beta}_1 S_{xx}$  again:

$$\hat{\beta}_1 S_{xy} = \hat{\beta}_1^2 S_{xx}$$

Since  $E[\hat{\beta}_1] = \beta_1$  and  $\text{Var}(\hat{\beta}_1) = \sigma^2/S_{xx}$ :

$$E[\hat{\beta}_1^2] = \text{Var}(\hat{\beta}_1) + (E[\hat{\beta}_1])^2 = \frac{\sigma^2}{S_{xx}} + \beta_1^2$$

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# The Final Result

Combining with the previous page:

$$E[SS_E] = E[S_{yy}] - E[\hat{\beta}_1 S_{xy}] = [\beta_1^2 S_{xx} + (n-1)\sigma^2] - [\sigma^2 + \beta_1^2 S_{xx}]$$

$$\boxed{E[SS_E] = (n-2)\sigma^2}$$

The  $\beta_1^2 S_{xx}$  terms cancel exactly — that cancellation is why  $SS_E$  isolates leftover error variance rather than variance explained by the true slope.

**Questions?**

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